

Diffraction $Z/\gamma^* \rightarrow \mu^+ \mu^-$ in CMS OpenData

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Work in progress / post Bonn hackathon v2

August, 2026

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Summary

Analysis strategy

Measurement definition

Physics target

The reconstructed fraction of single-diffractive $Z \rightarrow \mu\mu$ events in the complete event-and-track selection.

- *Expected* (naive) signal topology: reduced charged activity and a forward rapidity gap
- Dominant background: inclusive Drell–Yan (DY)
- Discriminants based on full event topology
- DY shapes are calibrated in an EGF sideband

Analysis sequence

1. Select $Z \rightarrow \mu\mu$ and quality tracks
2. Construct rapidity-gap observables
3. Reweight DY in data sidebands
4. Check out-of-fold closure in sideband slices
5. Fit the signal-enriched region
6. Extrapolate to the full selected-reconstruction range
7. **[not done]** Correct for muon & track efficiencies (+ unfold event level observables)

Samples and fiducial selection

Samples and event yields

Sample	ROOT files	Selected events	ML model use
Hard single-diffractive $Z \rightarrow \mu\mu$	6	94,555	5-fold held-out scoring
Inclusive DY $Z \rightarrow \mu\mu$	107	966,883	5-fold held-out scoring
2016H collision data	1,356	5,547,614	mean score from all folds

- Signal: DYToMuMu_pomflux_Pt-30.
- Background: ZToMuMu_M-50To120 (POWHEG+PYTHIA8).
- The NLO DY sample contains $\mathcal{O}(1\%)$ negative-weight events; event weights are neglected at this stage.
- Every ROOT file contributes to the plots and statistical templates.
- Model development is split at event level; every simulated event is scored only by an XGBoost fold that did not train on that event
- Of the 5,547,614 selected data events, 1,056,627 enter the restricted template-fit interval.

Input-file allocation and event usage

Full-sample use (every ROOT file enters plots and templates)

Sample	ROOT files	Input	Selected	Plots/templates
Single-diffractive MC	6	200,000	94,555	94,555
Inclusive DY MC	107	2,997,000	966,883	966,883
Data	1,356	174,035,164	5,547,614	5,547,614

Reference event-level XGBoost split (fold 0)

MC event split	Train	Validation	Test	Total
Single-diffractive MC	56,733	18,911	18,911	94,555
Inclusive DY MC	580,129	193,377	193,377	966,883

Final XGBoost templates use 100% of selected MC events; each MC event is evaluated only by its held-out fold model.

“Selected” denotes events passing the complete dimuon event selection. ROOT files are never reserved for training: each event is assigned to the train, validation, or test sample from its fixed event identifier.

Event and Track fiducial definition

Dimuon event selection

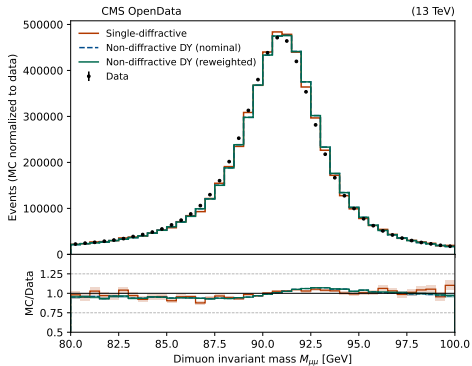
- IsoMu24 trigger.
- At least two loose muons.
- Leading two: $p_T^\mu > 25 \text{ GeV}$, $|\eta|^\mu < 2.4$.
- Opposite charge.
- $80 < m_{\mu\mu} < 100 \text{ GeV}$.

Tracks

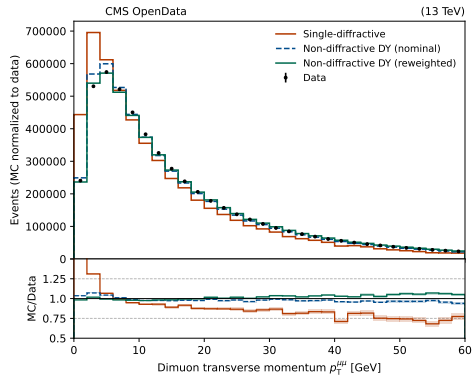
- $p_T > 0.5 \text{ GeV}$, $|\eta| < 2.5$, $\text{fromPV} \geq 2^*$.
- Nonzero charge; finite η , ϕ , and p_T .
- High-purity, no lost inner hits when available.
- $|d_{xy}| < 0.02 \text{ cm}$, $|d_z| < 0.10 \text{ cm}$.
- $|d_{xy}/\sigma|, |d_z/\sigma| < 3$ and normalized $\chi^2 < 5$ when available.
- Remove tracks within $\Delta R < 0.01$ of either selected muon.

* fromPV association quality: ≥ 1 = loose (JME), ≥ 2 = tight (isolation), and 3 = track used in the primary-vertex fit; 0 = no PV association. [\[CMSSW source\]](#)

Dimuon distributions

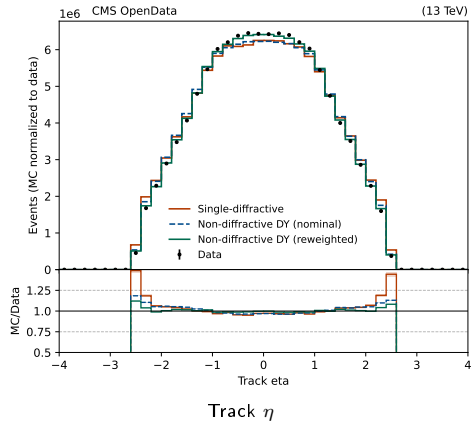
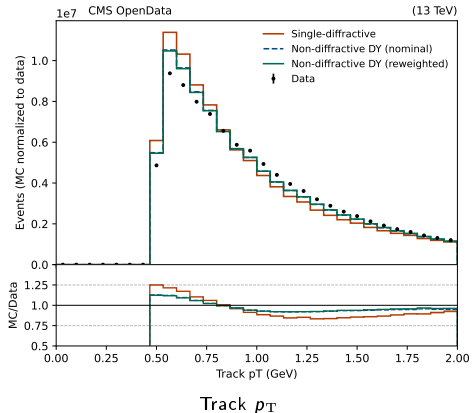


Dimuon invariant mass $M_{\mu\mu}$



Dimuon transverse momentum $p_T^{\mu\mu}$

Track distributions



Cutflows

Cutflow: full single-diffractive MC sample

Unit: events

Selection	Count	Step eff.	Cumulative
All input events	200,000	100.00%	100.00%
Trigger: IsoMu24	151,420	75.71%	75.71%
≥ 2 loose muons	136,286	90.01%	68.14%
≥ 2 loose muons with $p_T > 25$ GeV	106,256	77.97%	53.13%
≥ 2 selected muons with $ \eta < 2.4$	104,935	98.76%	52.47%
Opposite-sign leading pair	104,910	99.98%	52.45%
$80 < m(\mu\mu) < 100$ GeV	94,555	90.13%	47.28%

Unit: track candidates

Selection	Count	Step eff.	Cumulative
All track candidates	1,936,309	100.00%	100.00%
Finite p_T , η , and ϕ	1,936,309	100.00%	100.00%
Nonzero charge	1,936,309	100.00%	100.00%
$p_T > 0.5$ GeV	1,320,976	68.22%	68.22%
$ \eta < 2.5$	1,311,612	99.29%	67.74%
fromPV ≥ 2	1,311,612	100.00%	67.74%
Finite impact parameters and errors	1,311,408	99.98%	67.73%
$ dx < 0.02$ cm and $ dz < 0.1$ cm	787,476	60.05%	40.67%
$ dx /\sigma < 3$ and $ dz /\sigma < 3$	716,522	90.99%	37.00%
High-purity track	716,111	99.94%	36.98%
Muon overlap: $\Delta R \geq 0.01$	540,851	75.53%	27.93%

Cutflow: full inclusive DY MC sample

Unit: events

Selection	Count	Step eff.	Cumulative
All input events	2,997,000	100.00%	100.00%
Trigger: IsoMu24	1,831,091	61.10%	61.10%
>= 2 loose muons	1,640,185	89.57%	54.73%
>= 2 loose muons with $p_T > 25$ GeV	1,078,223	65.74%	35.98%
>= 2 selected muons with $ \eta < 2.4$	1,064,838	98.76%	35.53%
Opposite-sign leading pair	1,064,329	99.95%	35.51%
$80 < m(\mu\mu) < 100$ GeV	966,883	90.84%	32.26%

Unit: track candidates

Selection	Count	Step eff.	Cumulative
All track candidates	56,792,843	100.00%	100.00%
Finite p_T , η , and ϕ	56,792,843	100.00%	100.00%
Nonzero charge	56,792,843	100.00%	100.00%
$p_T > 0.5$ GeV	37,769,919	66.50%	66.50%
$ \eta < 2.5$	37,470,597	99.21%	65.98%
fromPV >= 2	37,470,597	100.00%	65.98%
Finite impact parameters and errors	37,468,663	99.99%	65.97%
$ dx < 0.02$ cm and $ dz < 0.1$ cm	25,562,274	68.22%	45.01%
$ dx /\sigma < 3$ and $ dz /\sigma < 3$	23,126,760	90.47%	40.72%
High-purity track	23,121,922	99.98%	40.71%
Muon overlap: $\Delta R \geq 0.01$	21,389,343	92.51%	37.66%

Cutflow: full data sample

Unit: events

Selection	Count	Step eff.	Cumulative
All input events	174,035,164	100.00%	100.00%
Trigger: IsoMu24	111,321,744	63.97%	63.97%
>= 2 loose muons	79,630,960	71.53%	45.76%
>= 2 loose muons with $p_T > 25$ GeV	6,587,765	8.27%	3.79%
>= 2 selected muons with $ \eta < 2.4$	6,519,083	98.96%	3.75%
Opposite-sign leading pair	6,437,294	98.75%	3.70%
$80 < m(\mu\mu) < 100$ GeV	5,547,614	86.18%	3.19%

Unit: track candidates

Selection	Count	Step eff.	Cumulative
All track candidates	293,541,603	100.00%	100.00%
Finite p_T , η , and ϕ	293,541,603	100.00%	100.00%
Nonzero charge	293,541,603	100.00%	100.00%
$p_T > 0.5$ GeV	205,241,763	69.92%	69.92%
$ \eta < 2.5$	203,885,911	99.34%	69.46%
fromPV >= 2	203,885,911	100.00%	69.46%
Finite impact parameters and errors	203,873,839	99.99%	69.45%
$ dx < 0.02$ cm and $ dz < 0.1$ cm	135,809,047	66.61%	46.27%
$ dx /\sigma < 3$ and $ dz /\sigma < 3$	122,690,081	90.34%	41.80%
High-purity track	122,652,378	99.97%	41.78%
Muon overlap: $\Delta R \geq 0.01$	112,930,701	92.07%	38.47%

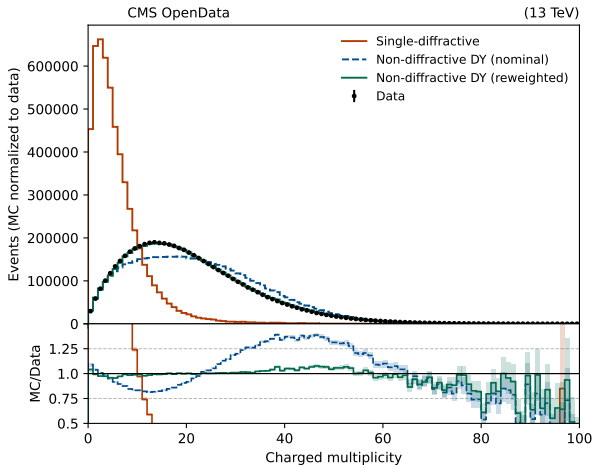
Observable definitions and distributions

Charged multiplicity

Definition

$$N_{\text{ch}} = \sum_{j \in \text{accepted}} 1.$$

Accepted tracks satisfy the full Track fiducial and quality selection.



Largest forward rapidity gap

Definition

For ordered accepted tracks

$$\eta_1 < \dots < \eta_N \text{ in } [\eta_{\min}, \eta_{\max}],$$

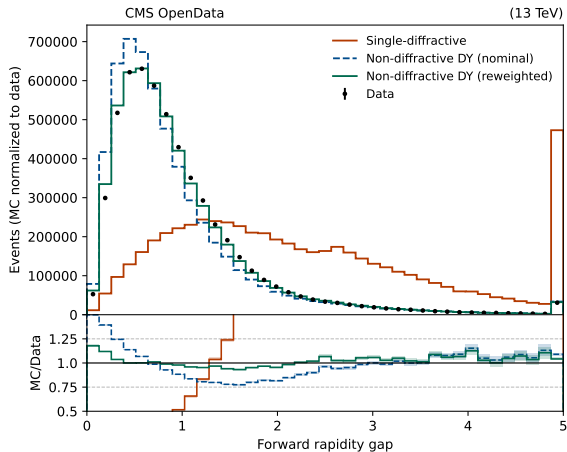
$$\Delta\eta_F = \max(\eta_1 - \eta_{\min}, \eta_{\max} - \eta_N).$$

For $N = 0$, define

$$\Delta\eta_F = L \equiv \eta_{\max} - \eta_{\min}.$$

Interpretation

Largest empty interval adjacent to either detector boundary.



Maximum rapidity gap

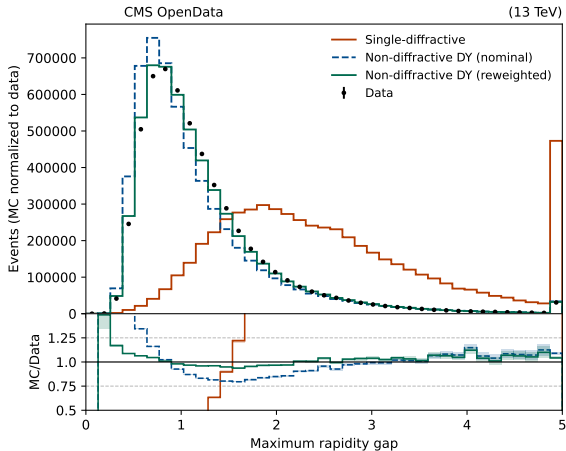
Definition

Add the detector boundaries as $\eta_0 = \eta_{\min}$ and $\eta_{N+1} = \eta_{\max}$, then

$$\Delta\eta_{\max} = \max_{0 \leq i \leq N} (\eta_{i+1} - \eta_i).$$

Interpretation

Widest empty interval in the accepted tracking volume.



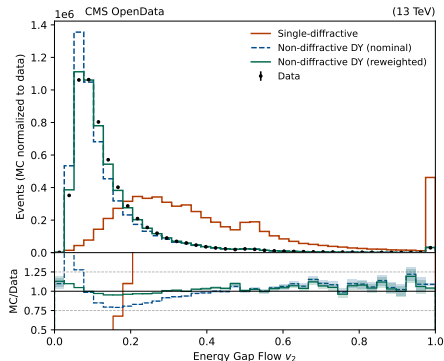
Energy Gap Flow v_2

Order the N accepted tracks as $\eta_1 < \dots < \eta_N$, set $\eta_0 = \eta_{\min}$ and $\eta_{N+1} = \eta_{\max}$, and define $\Delta\eta_i = \eta_{i+1} - \eta_i$ for $i = 0, \dots, N$ and $L = \eta_{\max} - \eta_{\min}$. For $i < j$, the tracks $k = i + 1, \dots, j$ lie between intervals i and j . Their activity separation weight is $q_0 = 0.15 \text{ GeV}$ (free param):

$$\text{EGF} = \frac{1}{L^2} \left[\sum_i (\Delta\eta_i)^2 + 2 \sum_{i < j} \Delta\eta_i \Delta\eta_j W_{ij} \right], \quad W_{ij} = \exp \left[-\frac{1}{q_0} \sum_{k=i+1}^j p_{T,k} \right].$$

Interpretation

The scalar $\sum p_T$ measures charged activity between empty η intervals. In the massless limit, the construction is invariant under longitudinal boosts within the fixed detector acceptance. Empty intervals with no activity between form one gap. Tracks with large p_T suppress their connection exponentially. Large EGF selects wide, low-activity regions characteristic of diffraction.



Control, validation, and fit regions

Energy-gap-flow boundaries

Region		EGF interval	Role
Control (CR)	slice	[0, 0.10]	Included in nominal training; defines the low-sideband reweighting variation.
Validation (VR)	slice	(0.10, 0.20]	Included in nominal training; provides an out-of-fold closure check and defines the high-sideband variation.
Fit / signal region		(0.20, 1.00]	Profile-likelihood template fit; not used to train the data/DY reweighter.
Full range	selected	[0, 1]	Denominator for the reconstructed-event fraction after template extrapolation.

Exclusive events (as a test)

Exclusive-event definition: $\gamma\gamma$ -fusion like dimuons

Selection

$$N_{\text{ch}} = 0,$$

$$A_\phi = 1 - \frac{|\Delta\phi_{\mu\mu}|}{\pi} < 0.02,$$

$$p_{\text{T}}^{\mu\mu} < 1.5 \text{ GeV}.$$

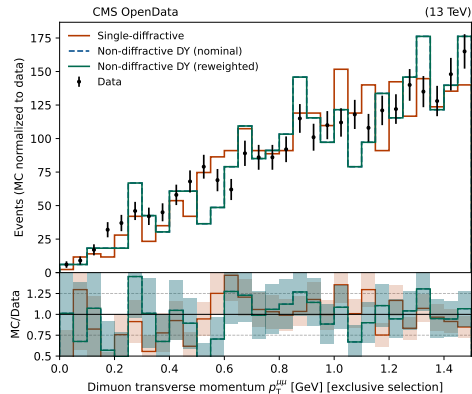
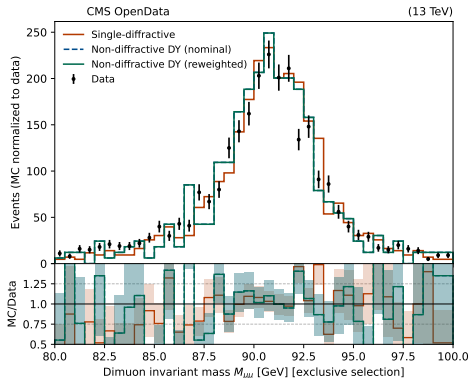
Here $|\Delta\phi_{\mu\mu}| \in [0, \pi]$ is the wrapped azimuthal separation. The zero-track requirement is evaluated after track-quality selection and muon-overlap removal.

Dimuon observables

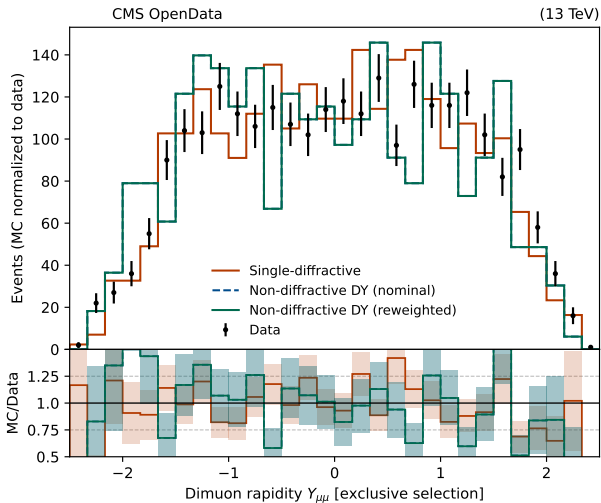
- dimuon invariant mass $M_{\mu\mu}$;
- dimuon transverse momentum $p_{\text{T}}^{\mu\mu}$;
- dimuon rapidity $Y_{\mu\mu}$.

All distributions show single-diffractive MC, non-diffractive DY (nominal), non-diffractive DY (reweighted), and data.

Exclusive events: dimuon mass and transverse momentum



Exclusive events: dimuon rapidity



Data-driven DY calibration

Multidimensional Data/MC DY reweighting

Classifier inputs

$p_T^{\mu\mu}$, $|Y_{\mu\mu}|$, N_{ch} , $\sum p_T$, and track count / scalar- p_T sum in eight η bins.

- XGBoost distinguishes data ($y = 1$) from DY ($y = 0$) in $\text{EGF} \leq 0.20$.
- The EGF fit variable and explicit gap observables are excluded from the input vector.
- Classes are balanced during training.

Density-ratio estimate

For classifier output $r(x) = P(y = 1 | x)$ and training-class priors $\pi_{\text{data}}, \pi_{\text{DY}}$,

$$\frac{p_{\text{data}}(x)}{p_{\text{DY}}(x)} = \frac{\pi_{\text{DY}}}{\pi_{\text{data}}} \frac{r(x)}{1 - r(x)}.$$

The balanced training priors cancel. Ratios are clipped to $[0.25, 4]$ and normalized to preserve the inclusive DY yield.

Diagnostics

Sideband AUC: 0.6063

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} : \quad 65,722 \rightarrow 61,115.$$

ESS is the equivalent unit-weight event count; its 7.0% decrease is caused by weight variance, not event removal.

Out-of-fold sideband weights

Each event is weighted by a classifier not trained on it

Use of the validation slice

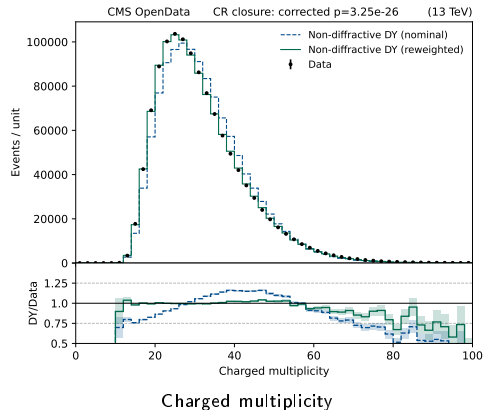
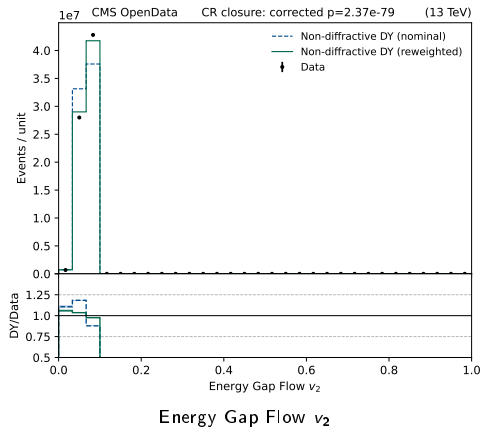
Both CR and VR enter the nominal training. The VR is therefore not an untouched validation sample; its closure plots use held-out predictions.

1. Split data and DY events with $\text{EGF} \leq 0.20$ into 3 folds while preserving their proportions.
2. For fold k , train the density-ratio classifier on all other folds with equal total data and DY weight, then evaluate it only on fold k .
3. Repeat for all folds. Every sideband DY weight then comes from a classifier that did not train on that event.

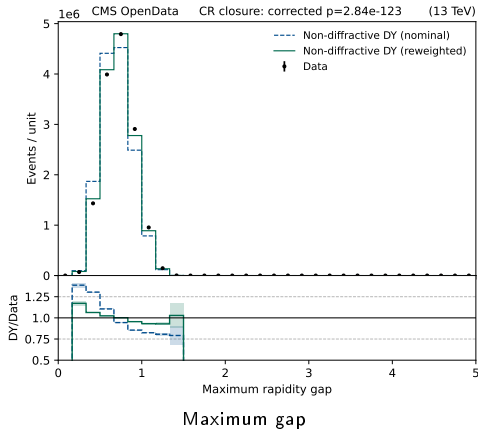
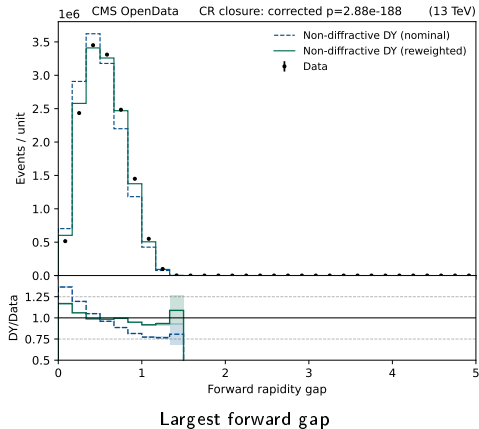
Application above the sideband

No event with $\text{EGF} > 0.20$ enters training. Fit-region DY uses the average prediction of the 3 sideband classifiers; the factors are normalized to change the DY shape but not its inclusive yield.

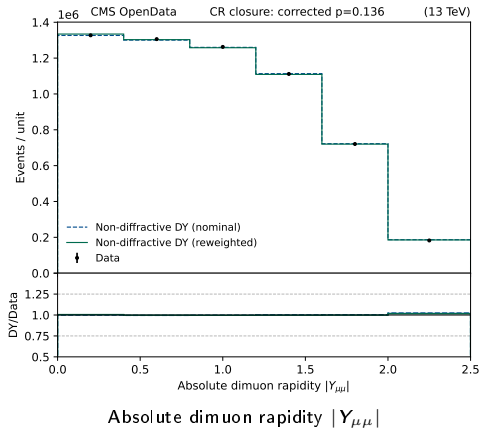
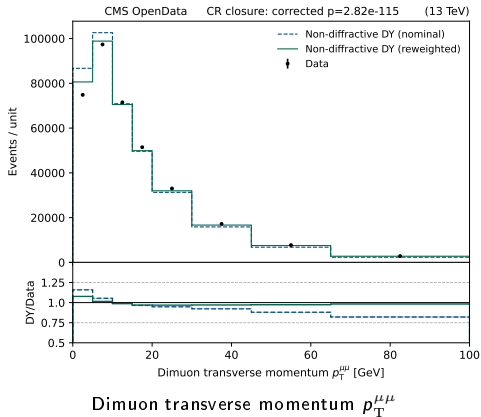
Control-region closure: fit variable and activity



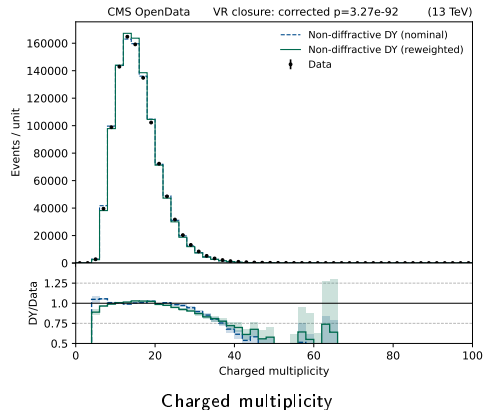
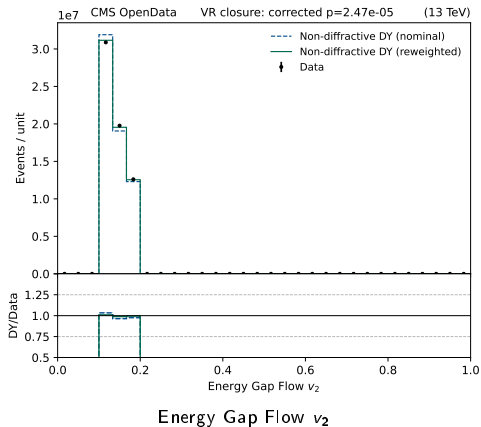
Control-region closure: rapidity gaps



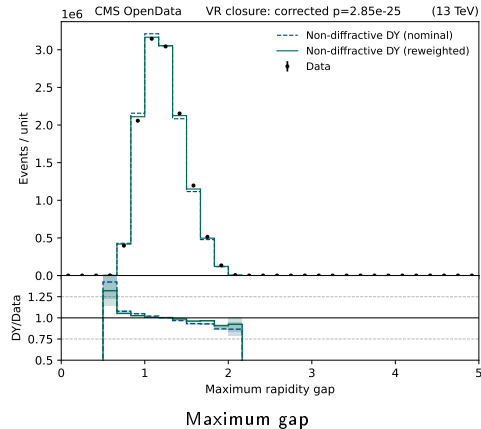
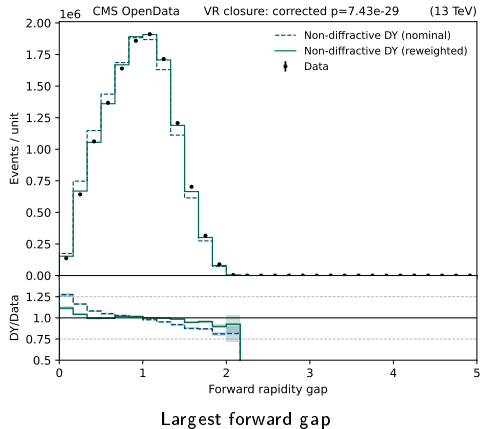
Control-region closure: dimuon kinematics



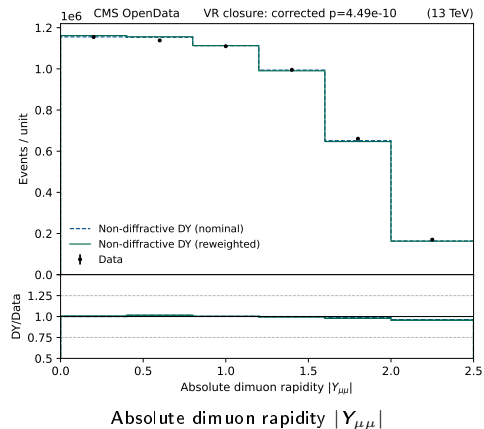
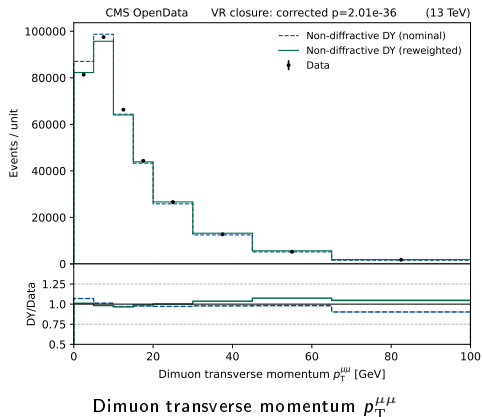
VR out-of-fold closure: fit variable and activity



VR out-of-fold closure: rapidity gaps



VR out-of-fold closure: dimuon kinematics



VR out-of-fold closure summary

Observable	VR shape p -value
N_{ch}	0.06152
Forward gap	0.02677
EGF	0.4968
$p_{\text{T}}^{\mu\mu}$	0.3349
$ Y_{\mu\mu} $	0.6444
Maximum gap	6.517×10^{-4}

Pearson χ^2 test with data and weighted-MC variances;
one degree of freedom is removed for normalization.

Interpretation

The small maximum-gap p -value shows that reweighted DY still misses this distribution. Because the VR belongs to the training sideband (but each event is scored out of fold), this checks model closure within the sideband—not extrapolation into the signal-fit region.

How closure is used

These p -values are not a pass/fail test. The low–high sideband difference and fold-to-fold spread become a correlated DY-shape uncertainty in the fit. The final model check is the joint post-fit goodness of fit.

CR-to-fit transfer uncertainty

Let $w_0(x)$ be the nominal correction trained in the full sideband. Train two additional corrections, $w_{\text{low}}(x)$ in $\text{EGF} \leq 0.10$ and $w_{\text{high}}(x)$ in $0.10 < \text{EGF} \leq 0.20$. For every DY event define

$$\delta_{\text{slice}}(x) = \max_{a \in \{\text{low}, \text{high}\}} \left| \ln \frac{w_a(x)}{w_0(x)} \right|, \quad \delta_{\text{model}}(x) = \text{Std}_k[\ln w_k(x)].$$

The first term measures how much the correction changes when it is learned at opposite ends of the sideband; the second measures the spread among the 3 nominal fold classifiers. They are combined as

$$\delta(x) = \min\left(\sqrt{\delta_{\text{slice}}^2(x) + \delta_{\text{model}}^2(x)}, \ln 2\right), \quad w_{\pm}(x) \propto w_0(x)e^{\pm\delta(x)}.$$

Each variation is renormalized to the nominal inclusive DY yield and then histogrammed into an EGF template. A Gaussian-constrained nuisance parameter morphs the background between the down, nominal, and up templates and is profiled in the signal fit.

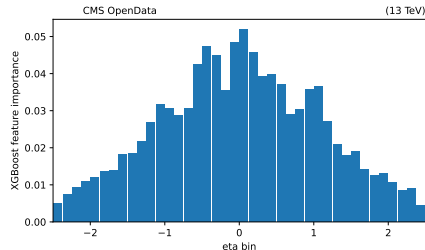
What this uncertainty covers

This is a shape uncertainty for classifier stability and extrapolation from the sideband. CR–VR closure remains a reported diagnostic and is not used to reject the result. Signal modelling and detector efficiencies require additional nuisance parameters.

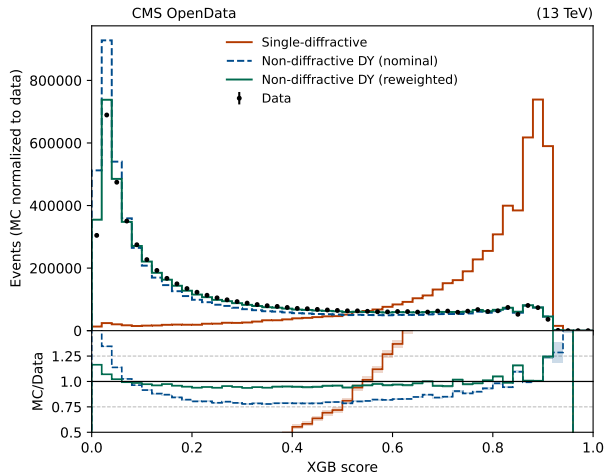
BDT discriminant (as a test)

XGBoost discriminant

- Forty η bins containing scalar track- p_T sums.
- Kinematic-equalization weights match the single-diffractive and inclusive DY training samples in the joint $(p_T^{\mu\mu}, |Y_{\mu\mu}|)$ distribution.
- For each cross-fit classifier, the kinematic-equalization map is fitted only on its training events. These weights are training-only, the analysis DY template instead carries the data-driven DY reweighting.



BDT score



Track-level Transformer (as a test)

Transformer

Per-track inputs

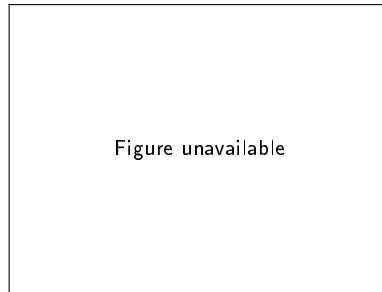
For every charged track passing the analysis track selection,

$$\mathbf{x}_i = \left(\frac{p_{T,i}}{5 \text{ GeV}}, \frac{\eta_i}{\eta_{\max}}, \sin \phi_i, \cos \phi_i \right).$$

The sine and cosine representation avoids the discontinuity at $\phi = \pm\pi$. No event-level gap observable is supplied explicitly.

Network

Each track is embedded into 64 components. Three blocks of four-head self-attention plus a 128-component MLP are followed by a binary MLP classifier. PyTorch scaled-dot-product attention uses the FlashAttention kernel when supported by the GPU.



Training loss and validation AUC

Symmetries and training

Without positional encoding, the score is invariant under track reordering. Training-only kinematic-equalization weights match signal and inclusive DY in $(p_T^{\mu\mu}, |Y_{\mu\mu}|)$; each map is learned only from that cross-fit model's training events. Analysis DY templates instead carry the data-driven DY reweighting.

Charged-track Transformer score

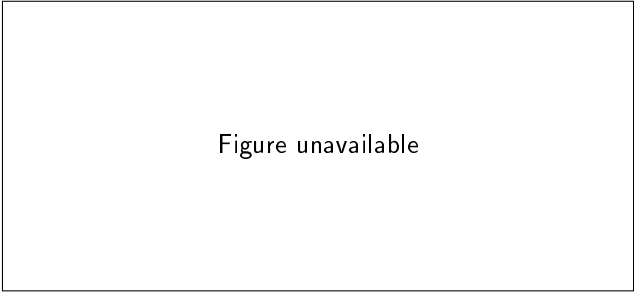
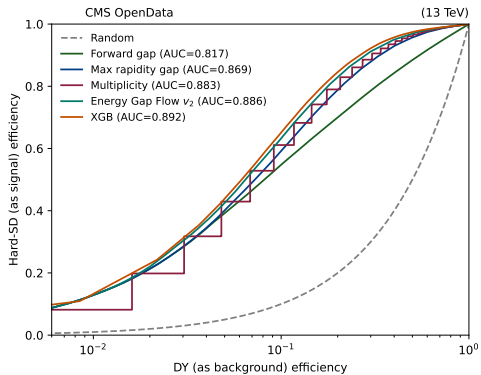


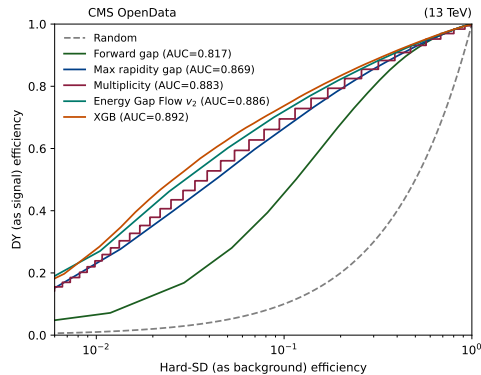
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ROC

ROC curves



Hard single diffraction treated as signal



Inclusive DY treated as signal (inverted definition !)

Choosing the final discriminant or observable

Ranking metric

The ultimate ranking (within simulation world ...) should be based on the expected profile-likelihood sensitivity, not from ROC/AUC alone.

1. Produce an expected limit for every candidate observable using the same event selection, bin-statistics requirements, and nuisance model.
2. Rank the observables by the median expected 95% CL_s upper limit (or expected precision in the measurement regime).
3. Test multidimensional combinations, including SBI with a parameterized estimator of the event-level likelihood ratio. A histogram of a fixed XGB or Transformer score is still ordinary template inference, not full SBI.

Profile-likelihood signal extraction

Conditional multinomial likelihood

Definitions

For bins $i = 1, \dots, B$, n_i is the observed count; $s_i(\gamma_s, \theta)$ and $b_i(\gamma_b, \theta)$ are the unit-normalized single-diffractive and DY probabilities, with $\sum_i s_i = \sum_i b_i = 1$. The vector θ contains all correlated template-shape nuisance parameters and $\gamma_{s,b}$ are Barlow–Beeston MC-statistical scale factors. Define $N \equiv N_{\text{data}}^{\text{fit}} = \sum_i n_i$ and $D \equiv N_{\text{data}}^{\text{sel}}$: the observed data totals in the fitted score interval and the complete selected interval, respectively. Both N and D are fixed.

Fit-region composition

The only unconstrained composition parameter is

$$0 \leq f^{\text{fit}} \leq 1.$$

At fixed total N , the bin probabilities and derived component counts are

$$p_i(f^{\text{fit}}, \gamma, \theta) = f^{\text{fit}} s_i(\gamma_s, \theta) + (1 - f^{\text{fit}}) b_i(\gamma_b, \theta),$$

$$N_s^{\text{fit}} = f^{\text{fit}} N, \quad N_b^{\text{fit}} = (1 - f^{\text{fit}}) N.$$

Hence $N_s^{\text{fit}} + N_b^{\text{fit}} = N$ exactly.

Effective-count Barlow–Beeston

$$\begin{aligned} \mathcal{L}_{\text{cond}} = & \text{Mult}(\{n_i\} \mid N, \{p_i\}) \prod_k G(\theta_k; 0, 1) \\ & \times \prod_{x \in \{s, b\}} \prod_i \text{Pois}(m_{x,i}^{\text{eff}} \mid \gamma_{x,i} m_{x,i}^{\text{eff}}). \end{aligned}$$

For weighted MC, $m_{x,i}^{\text{eff}} = (\sum w)^2 / \sum w^2$ and the auxiliary contribution (constant removed) is $m_{x,i}^{\text{eff}}(\gamma_{x,i} - 1 - \ln \gamma_{x,i})$. Each component bin is scaled by $\gamma_{x,i}$ and then renormalized inside the multinomial. This is exact for equal positive weights and is a moment-matched effective-count extension for unequal nonnegative weights.

Numerical inference: JAX and bounded L-BFGS-B

Likelihood and derivatives in JAX

The code evaluates the conditional likelihood in JAX with 64-bit arithmetic and compiles

$$(-\ln \mathcal{L}, \nabla_{\phi}[-\ln \mathcal{L}]) = \text{jit}(\text{value_and_grad})(\phi).$$

JAX supplies the objective and exact automatic derivatives for L-BFGS-B.

Bounded profiling with SciPy L-BFGS-B

SciPy drives the optimization using the JAX value and gradient:

- $0 \leq f^{\text{fit}} \leq 1$
- positive Barlow–Beeston scale factors
- $-5 \leq \theta_k \leq 5$ for Gaussian shape nuisances
- four global starts in f^{fit} , retaining the converged solution with the smallest negative log likelihood.

At every tested POI value, the POI is fixed and all conditional-fit nuisances are reoptimized and the nearest cached profile point initializes the next minimization.

Limit calculation

Brent root finding solves $q = 1$ for the approximate 68% profile interval and $\text{CL}_s = 0.05$ for limits. The signal fit-range-fraction nuisance is separately bounded and profiled at every selected-event fraction.

From the fitted fraction to the selected-event fraction

Parameter of interest

Let N_s^{sel} be the single-diffractive yield and D the total observed data count in the full selected reconstructed interval. The reported quantity is

$$\mu \equiv f_{\text{diff}}^{\text{sel}} = \frac{N_s^{\text{sel}}}{D}.$$

Here $A_s(\eta_A)$ is the fraction of selected signal entering the fitted score interval. The constrained latent parameter η_A has a finite profiling range; A_s follows the bounded split-logit response defined below.

Therefore

$$N_s^{\text{fit}} = \mu D A_s(\eta_A).$$

Combining this with $N_s^{\text{fit}} = f^{\text{fit}} N$ gives

$$f^{\text{fit}}(\mu, \eta_A) = \frac{\mu D A_s(\eta_A)}{N}.$$

Profiled fit-range fraction

The nuisance η_A is reprofiled for every tested μ ; $A_s(\eta_A)$ itself is not Gaussian. The likelihood is

$$\mathcal{L}(\mu, \gamma, \theta, \eta_A) = \mathcal{L}_{\text{cond}}\left(f^{\text{fit}}(\mu, \eta_A), \gamma, \theta\right) G(\eta_A; 0, 1).$$

The fitted and selected fractions differ because the low-observable region is excluded and $N \neq D$.

The fit uses shape information only

There is no independent signal or DY yield parameter. The observed fit-region total is conditioned upon; discrimination comes only from the binned template shape.

Limit construction for the selected-event fraction

Profile-likelihood statistic

For $\mu \equiv f_{\text{diff}}^{\text{sel}} \geq 0$ and all nuisance parameters $\psi = (\gamma, \theta, \eta_A)$,

$$\tilde{q}_\mu = \begin{cases} -2 \ln \frac{\mathcal{L}(\mu, \hat{\psi}_\mu)}{\mathcal{L}(\hat{\mu}, \hat{\psi})}, & \mu \geq \hat{\mu}, \\ 0, & \mu < \hat{\mu}. \end{cases}$$

Here $\mathcal{L}$ is the constrained likelihood defined previously, $\hat{\mu} \geq 0$ is the global best fit and $\hat{\psi}$ its fitted nuisance vector. $\hat{\psi}_\mu$ is the conditional nuisance fit at fixed μ ; $\tilde{q}_\mu$ is set to zero below the best fit. The constrained best fit and the physical boundary at $\mu = 0$ define the bounded one-sided statistic.

Bounded asymptotic CL_s

Let $q_o = \tilde{q}_{\mu, \text{obs}}$ and $q_A = \tilde{q}_{\mu, A}$, and define

$$t_\mu = \begin{cases} \sqrt{q_o} - \sqrt{q_A}, & q_o \leq q_A, \\ (q_o - q_A)/(2\sqrt{q_A}), & q_o > q_A. \end{cases}$$

The second branch is required for downward fluctuations at the physical boundary. With Φ the standard-normal CDF and $\text{CL}_b \equiv 1 - p_b$,

$$\text{CL}_{s+b}(\mu) = \Phi[-(t_\mu + \sqrt{q_A})],$$

$$\text{CL}_b(\mu) = \Phi(-t_\mu),$$

$$\text{CL}_s(\mu) = \text{CL}_{s+b}/\text{CL}_b,$$

$$\text{CL}_s(\mu_{95}^{\text{obs}}) = 0.05.$$

The root is searched for $\mu \geq \hat{\mu}$ and all nuisance parameters are reprofiled at every tested point.

Background-only expected bands

Asimov construction

At the same fixed observed total, the background-only Asimov counts are

$$n_{i,A} = N_{\text{data}}^{\text{fit}} b_i^0, \quad \sum_i b_i^0 = 1.$$

Here b_i^0 is the nominal background probability in bin i and $n_{i,A}$ is the corresponding Asimov count. For a background-only fluctuation quantile $n \in \{-2, -1, 0, +1, +2\}$,

$$\text{CL}_s^{(n)}(\mu) = \frac{1 - \Phi\left(\sqrt{\tilde{q}_{\mu,A}} - n\right)}{\Phi(n)}.$$

Each expected limit is obtained by directly solving $\text{CL}_s^{(n)}(\mu_{95}^{(n)}) = 0.05$, with nonlinear nuisance profiling repeated at every μ . The median is $n = 0$; the 68% and 95% bands use $n = \pm 1$ and $n = \pm 2$.

How to interpret these bands

These are direct asymptotic inversions, not pseudoexperiment-derived expected bands. The fit GOF can separately use conditional multinomial toys with fixed template auxiliary observations.

Template extrapolation and reported quantity

Fit-to-selected template fraction

The fit uses $N_s^{\text{fit}} = A_s(\eta_A) N_s^{\text{sel}}$, where η_A has a standard-normal constraint. For nominal fit-range fraction A_0 and total absolute uncertainty δA , define

$$A_{\pm} = \text{clip}(A_0 \pm \delta A, \epsilon_A, 1 - \epsilon_A), \quad \ell_0 = \text{logit}(A_0), \quad \ell_{\pm} = \text{logit}(A_{\pm}).$$

The configured split-logit-normal response is

$$\ell(\eta_A) = \begin{cases} \ell_0 + \eta_A(\ell_+ - \ell_0), & \eta_A \geq 0, \\ \ell_0 + \eta_A(\ell_0 - \ell_-), & \eta_A < 0, \end{cases} \quad A_s(\eta_A) = \text{logistic}[\ell(\eta_A)].$$

Hence $A_s(0) = A_0$, $A_s(\pm 1) = A_{\pm}$, and $0 < A_s < 1$ without a clipped plateau. The YAML configuration specifies the response; δA combines finite-MC and configured systematic uncertainties in quadrature.

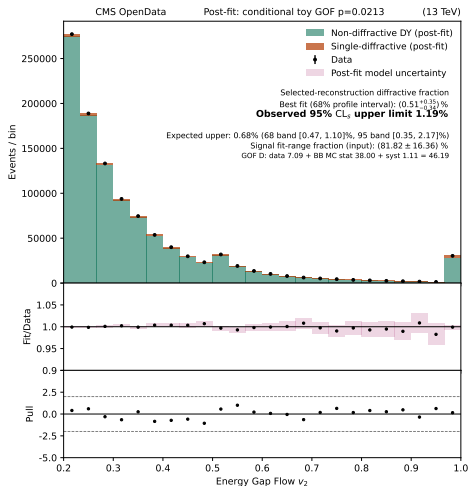
Reported upper limit

Since $D = N_{\text{data}}^{\text{sel}}$ is fixed,

$$\mu_{95} = f_{\text{diff}, 95}^{\text{sel}} = \frac{N_{s, 95}^{\text{sel}}}{N_{\text{data}}^{\text{sel}}}.$$

This is a conditional selected-reconstruction fraction. A physical fiducial cross section would additionally require validated luminosity, detector efficiencies, and particle-level acceptance, so none is quoted.

Signal extraction fit and result



Goodness-of-fit

Conditional toy MC GOF: $p = 0.0213$.

$$D_{\text{data}} = 7.09$$

$$D_{\text{BB}} = 38.00$$

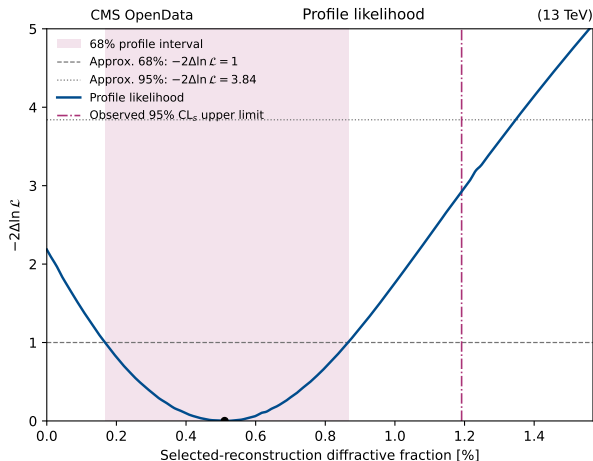
$$D_{\text{syst}} = 1.11$$

$$D_{\text{total}} = 46.19.$$

The low p -value indicates residual tension. The deviance D is dominated by finite-MC template shifts (Barlow–Beeston) rather than the post-fit data residual.

- Inspect the largest BB pulls and per-bin effective counts
- Increase MC sample stats (or merge bins)
- Repeat

Profile likelihood for the diffractive fraction



In the limit-setting regime the vertical line is the observed 95% CL_s upper limit; a significant measurement would instead use the two-sided profile interval.

Nuisance-impact plots details

Local covariance

The fitted parameter vector is $\phi = (f^{\text{fit}}, \gamma_s, \gamma_b, \theta)$. At the observed best fit,

$$H_{ab} = \left. \frac{\partial^2 [-\ln \mathcal{L}_{\text{cond}}]}{\partial \phi_a \partial \phi_b} \right|_{\hat{\phi}}, \quad C = H^+,$$

where H^+ is the numerically regularized inverse on directions with positive curvature. Derivatives respect the physical parameter bounds.

Local one-standard-deviation impact

For nuisance j , $\sigma_j^{\text{post}} = \sqrt{C_{jj}}$. With $\mu = f_{\text{diff}}^{\text{sel}}$ and $\mu = N f^{\text{fit}} / (A_0 D)$ at the nominal fit-range fraction, the plotted local response is

$$\Delta\mu_j^{\pm} = \pm \frac{N}{A_0 D} \frac{C_{fj}}{\sigma_j^{\text{post}}}.$$

Rows included

- one Barlow–Beeston MC-statistics nuisance for every retained signal and DY template bin
- every correlated DY template-shape nuisance
- the signal fit-range-fraction nuisance

The fit-range-fraction row uses the exact configured shifts

$$\Delta\mu_A^{\pm} = \hat{\mu} \frac{A_0}{A_{\pm}} - \hat{\mu}$$

rather than the conditional-fit covariance, and rows are ranked by $\max(|\Delta\mu^+|, |\Delta\mu^-|)$.

Impact plots: pull and post-fit uncertainty

Gaussian nuisance θ_j

The standard-normal constraint has $\theta_{j,0} = 0$ and $\sigma_{j,\text{pre}} = 1$. The plotted point and half-width are

$$p_j = \hat{\theta}_j, \quad w_j = \sigma_j^{\text{post}} = \sqrt{C_{jj}}.$$

Thus the horizontal interval is

$$[p_j - w_j, p_j + w_j]$$

in units of the pre-fit standard deviation.

Reading the left panel

$w < 1$ means the fit constrains the nuisance; $w \simeq 1$ means little information beyond its pre-fit constraint. A large $|p|$ indicates tension with the nominal template, not necessarily a large signal impact.

Barlow–Beeston bin i

For MC-event weights w_e in bin i ,

$$n_{i,\text{eff}} = \frac{(\sum_{e \in i} w_e)^2}{\sum_{e \in i} w_e^2},$$

$$P_i(\gamma) = \text{sgn}(\gamma - 1) \sqrt{2n_{i,\text{eff}}(\gamma - 1 - \ln \gamma)},$$

$$p_i = P_i(\hat{\gamma}_i), \quad w_i = |P'_i(\hat{\gamma}_i)| \sqrt{C_{\gamma_i \gamma_i}}.$$

Thus both the point and its propagated local Hessian width are in the same exact signed Poisson-deviance coordinate.

Impact plots: response of the fitted fraction

Conditional-fit nuisance j

For a local $\pm 1\sigma_j^{\text{post}}$ displacement,

$$\sigma_j^{\text{post}} = \sqrt{C_{jj}}, \quad \delta f_j^{\pm} = \pm \frac{C_{fj}}{\sqrt{C_{jj}}}.$$

Since $\mu = Nf^{\text{fit}}/(A_0D)$ at the nominal fit-range fraction,

$$\Delta\mu_j^{\pm} = \frac{N}{A_0D} \delta f_j^{\pm}, \quad I_j^{\pm} = 100 \Delta\mu_j^{\pm}.$$

The displayed $I_j^{\pm}$ are absolute percentage-point changes.

Note!

Nuisances are correlated through C . The plot ranks individual local responses; it is neither a decomposition of the total uncertainty nor a nuisance-freeze/reprofile change in the 95% CL_s upper limit. At an active parameter bound, the symmetric local-covariance interpretation is only approximate.

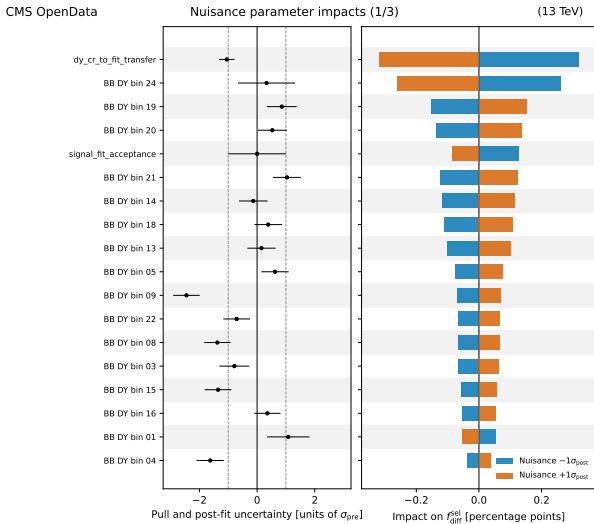
Fit-range-fraction nuisance

This row uses the exact bounded one-standard-deviation shifts rather than the conditional-fit covariance:

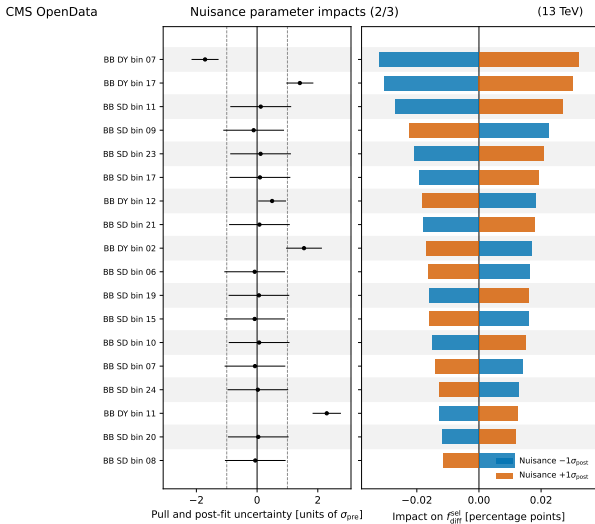
$$I_A^{\pm} = 100 \hat{\mu} \left(\frac{A_0}{A_{\pm}} - 1 \right), \quad A_{\pm} = A_s(\pm 1).$$

Thus $I = +0.10$ moves, for example, $\mu = 1.00\%$ to 1.10% .

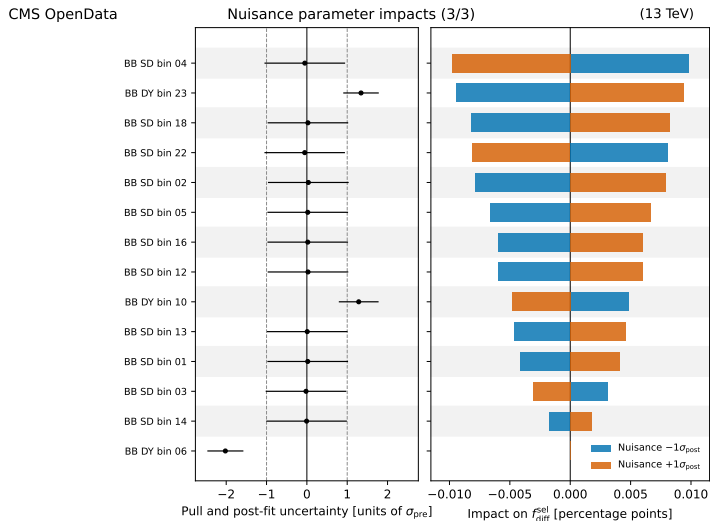
Nuisance-parameter impacts (1/3)



Nuisance-parameter impacts (2/3)



Nuisance-parameter impacts (3/3)



Summary

Systematic model and required additions

Implemented

- analyzed the complete 2016H data sample
- per-sample, per-bin effective-count Barlow–Beeston terms
- data-driven DY CR-to-fit shape transfer
- signal fit-to-selected template-fraction uncertainty
- bounded $\tilde{q}_\mu$ CL_s and directly inverted bands
- joint post-fit GOF with optional conditional toy calibration

Required

- signal shape/acceptance variations (e.g. via gen-level transform with Transformers)
- muon trigger, reconstruction, and ID efficiencies / scale factors
- track efficiency/fake-rate/material variations
- pileup and residual background composition
- luminosity for an absolute cross section
- x-check with alternative reweighters and sideband variations
- x-check with alternative diffraction definitions (or hi-dim SBI)

Summary

- Energy Gap Flow v_2 used as the diffraction *defining* observable.
- A hi-dim data/DY classifier calibrates the background using only $\text{EGF} \leq 0.20$.
- Sideband-slice disagreement is included in the fit as a correlated CR-to-fit shape nuisance.
- CR–VR closure is shown as a diagnostic and is not used to reject the result.

Open topics

- Larger (signal) MC stats needed for templates
- Track quality cuts require more analysis, efficiencies / scale factors
- Fiducial differential x-sections $\Rightarrow$ Muon efficiencies (trigger, ID, isolation scale factors), Track efficiencies + (neural)-unfolding of event level hadronic observables, signal shape systematics etc.
- Track efficiencies possibly with D^* mesons: [CMS-DP-2018-050 pion-tracking efficiency](#)
- (Possibly) Multidimensional efficiency correction with [DeepEfficiency \(MM\)](#), [arXiv:1809.06101](#), requires normal MC and/or data-driven input

Log-scale histogram appendix

Dimuon distributions: logarithmic scale

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Dimuon invariant mass $M_{\mu\mu}$

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Dimuon transverse momentum $p_T^{\mu\mu}$

Tracks: logarithmic scale

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Track transverse momentum

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Track pseudorapidity

Rapidity-gap observables: logarithmic scale

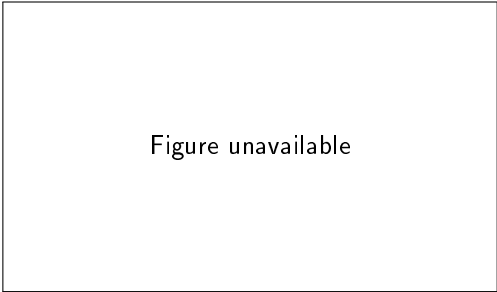


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Largest forward rapidity gap

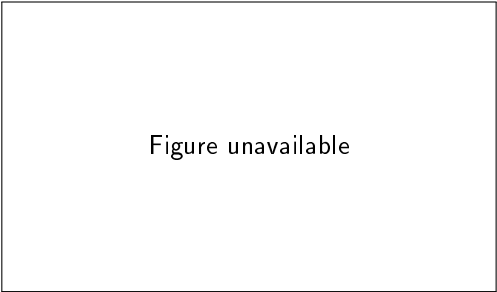


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Maximum rapidity gap

Activity observables: logarithmic scale

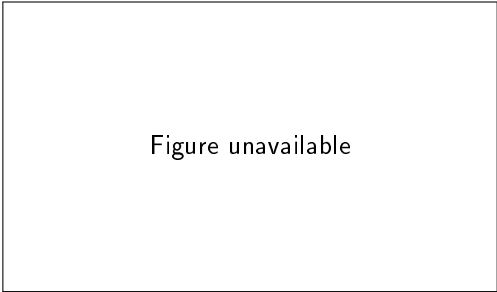


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Energy Gap Flow v_2

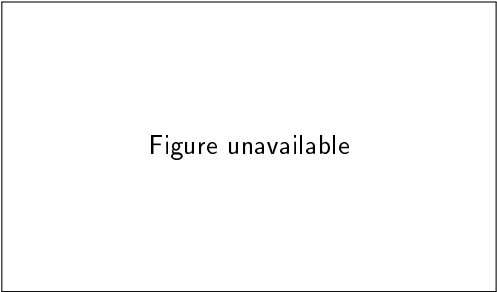


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Charged multiplicity

BDT score: logarithmic scale

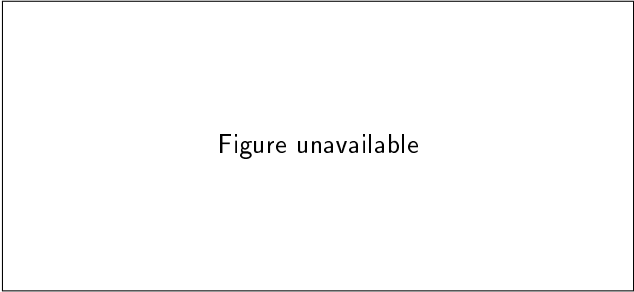


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Transformer score: logarithmic scale

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Exclusive mass and transverse momentum: logarithmic scale

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Exclusive rapidity: logarithmic scale

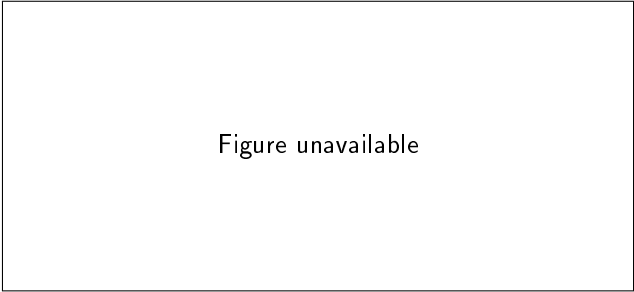


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Control region: activity observables, logarithmic scale

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Control region: rapidity gaps, logarithmic scale

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Control region: dimuon kinematics, logarithmic scale

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Validation region: activity observables, logarithmic scale

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Validation region: rapidity gaps, logarithmic scale

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Validation region: dimuon kinematics, logarithmic scale

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